

Top 5 Must-Have Features

Same logic as Revenue Agent: pick the smallest set that forms one complete, demoable loop — *issue detected* → *triaged* → *technician briefed* → *resolved* → *status synced back to Front Desk/Revenue*.

1. **Work Order Intake & Priority Triage** — the entry point; nothing works without it.
 2. **Room Status Sync** — proves the multi-agent integration (feeds Front Desk directly).
 3. **Equipment Failure Prediction** — this is the PS's literal ask ("predictive maintenance to identify potential equipment or facility failures before they affect guests").
 4. **Cost Feed to Revenue Agent** — makes your Revenue Agent's Net RevPAR feature *real* instead of hypothetical; strong cross-agent story for judges.
 5. **Contextual Repair Briefs** — cheap to build (just templated generation over existing history data), but it's the feature that makes the agent feel *intelligent* rather than a ticketing system with extra steps.
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2 Recommended Differentiators

Feature	Why It Wins	Build Effort
Computer Vision Triage	Best live-demo moment in the entire spec — "snap a photo, agent identifies the part, technician arrives with the right part first trip." Judges remember visual demos. Directly answers the PS's "beyond displaying data" bar.	Medium-High
Micro-Leak & Phantom Load Detection	Cheapest-to-build, highest-credibility differentiator — pure rule-based anomaly detection on mock data, but the <i>story</i> ("a running toilet in a vacant room costs thousands and nobody catches it") is exactly the kind of concrete, relatable insight that lands with judges. Also pairs naturally with your Cost Feed to Revenue Agent.	Low-Medium

These two together give you one flashy, visual differentiator (CV Triage) and one cheap, narratively sharp one (Leak Detection) — good balance of demo risk vs. payoff.

Add-Ons If Time Permits (in priority order)

1. **Weather-Triggered Pre-Emptive Protocols** — cheap, demoable, and ties nicely into "predictive" positioning without much build risk.
2. **Silent-Defect Detection** — strong integration story with Front Desk's Pre-Arrival Friction Alerts, moderate effort.
3. **Dynamic Grid Load Shedding** — good synergy with Revenue Agent, but only worth it if Micro-Leak Detection is already done and stable.

4. **Cascading Failure Alerts** — nice technical depth, but only if you have spare time — the dependency graph adds real build complexity for a feature that's hard to demo convincingly in a short window.

Cut entirely for the hackathon: Time-Bound Smart Access Provisioning (security-sensitive and hard to fake credibly), Gamified Sensor Network Reporting (needs a real guest-app surface, not just backend — bigger scope than it appears), Parts & Consumables Depletion Forecasting (low demo payoff for the effort).